UNIVERSITÀ DEGLI STUDI DI UDINE

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INTERNET OF THINGS PROJECT

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MINIMAL MIRROR

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UNIVERSITÀ DEGLI STUDI DI UDINE

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1 INTRODUCTION

To carry out the Exam Project in question, we decided to design and build a Smart Mirror, an intelligent, modular and open-source mirror powered by a Raspberry Pi 3 B +.
What makes a mirror "smart" is the ability to represent on it some of the essential information that we normally also find on our smartphones in the form of a widget.
More precisely, it is the set of all those graphic elements that communicate information such as the time, calendar, weather, news, etc. in a clear and simple way.
Therefore, we thought it would be useful and beneficial to design a mirror that has such capabilities.
Our smart mirror will take the name of MinimalMirror.

2 APPROACH AND MOTIVATIONS

2.1 APPROACH

- We initially focused on the overall structure of the project.
- First of all, the realization of our project was divided into two macro-phases:
- **Hardware phase:** purchase and assembly of components;
- **Software phase:** installation and configuration of the Raspberry Pi OS, installation of the Magic Mirror, implementation of the various modules chosen.

Our work was structured in the following phases:

1. Discussion of ideas
2. Choice of design tools and development methods
3. Definition of the design phases
4. Software analysis, design, implementation

In the first phase of work, we both proposed ideas on how to implement a "smart" mirror that could be useful and what main information to show to the user during its use.

Subsequently, we moved on to the choice of design tools: Github for the design and documentation of the project itself and Slack for communication and information sharing.

Given our choice to carry out a practical project, and not a theoretical one, we also carried out a cost analysis, to evaluate well the costs necessary for the realization of the project.
After this phase, we purchased the necessary components through Amazon (to speed up delivery times) and physical stores.

While waiting for the components, we focused on the second phase of the project, the software phase: on how to install the Raspberry Pi OS, how to install and configure the Magic Mirror and the various modules.

As soon as the components arrived, we created a ladder with the construction phases of the mirror, to carry them out in the correct order.

After building the entire mirror, we connected, configured and tested it with the software part, that is the modules.

Finally, we have drawn up the following report, trying to describe in an exhaustive manner the main design and implementation choices and the phases through which we have passed to carry out the project.

2.2 MOTIVATIONS

Our choice to carry out a practical project is due to our strong desire to realize a real product with practical use in everyday life.
We have tried to include technologies also covered in class to test our knowledge about it.
We have set ourselves the goal of creating a "smart" mirror as close as possible to the minimalist current of thought.

3 CONTRIBUTION OF MEMBERS

TASK	MEMBER
Hardware	Muhamed Kouate, Aleksej Petricig
Software	Muhamed Kouate
Relazione	Muhamed Kouate, Aleksej Petricig

4 HARDWARE

The hardware part of our project includes the actual construction of our MinimalMirror.

4.1 SEARCH OF COMPONENTS

During this phase we did research online, on where it was more convenient to buy the components we needed.
The components we use were mainly purchased on the amazon.it e-commerce platform, a choice due to its competitive prices and fast delivery times.
However, other products, given their particular nature, were found in their respective physical stores:

- IKEA: Ribba Frame 50cmx70cm;
- VETRERIA VENTURINI LUCA: Glass made to measure 50cmx70cm;

4.2 ANALYSIS OF COSTS

At this stage we calculated the expenses that we would have to incur for the purchase of the components necessary for the construction of the mirror.

COMPONENT	PRICE (€)	LINK
Controller EDP	36,07	https://www.amazon.it/dp/B07PVBCY3K/?coliid=I2FYNB2QF8FAH4&colid=Q5E9HQTSSTDY&psc=1&ref_=lv_ov_lig_dp_it
Kit Raspberry Pi 3 B+	16,78	https://www.amazon.it/UCreate-Raspberry-Pi-Desktop-Starter/dp/B07BNQ2TWW/ref=sr_1_8?__mk_it_IT=%C3%85M%C3%85%C5%BD%C3%95%C3%91&dchild=1&keywords=rasperry+pi+3+b%2B&qid=1594374031&s=pc&sr=1-8
Alim. Ktec 12V 3A	64,00	https://www.amazon.it/dp/B00JINQPJA/?coliid=I149OKSTOWB3J5&colid=Q5E9HQTSSTDY&psc=1&ref_=lv_ov_lig_dp_it
Black frame IKEA (50x70cm)	15,00	https://www.ikea.com/it/it/p/ribba-cornice-nero-50268874/
Transparent glass (50x70cm)	5,00	http://www.vetreriaventurini.it/

COMPONENT	PRICE (€)	LINK
Rhodesy adherent film, silver mirror (60x200cm)	25,99	https://www.amazon.it/gp/product/B07R4Y6HHY/ref=ppx_yo_dt_b_asin_title_o00_s00?ie=UTF8&psc=1
TOTAL	162,84	-

4.3 MATERIALS

2 plywood sheets

1 Cutter

1 Hot glue gun

1 transparent glass (50x70 cm)

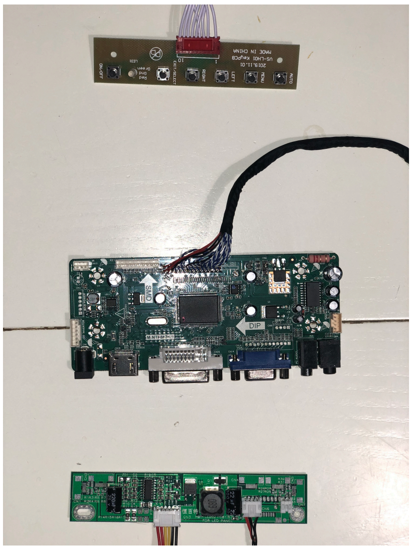
1 Frame IKEA - RIBBA
(50x70 cm)

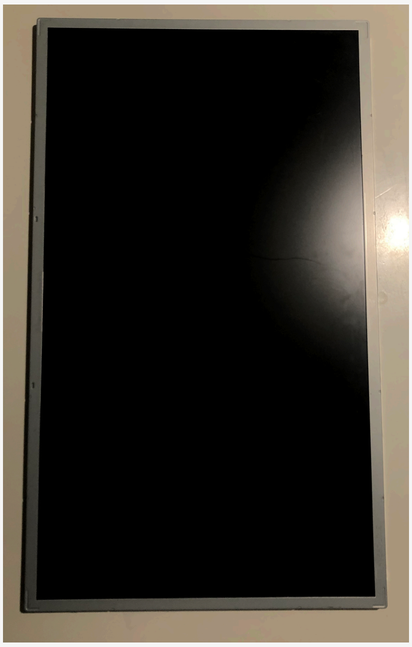
1 reflective film

1 paper tape

6 screws (3x16 mm)

4.4 COMPONENTS

COMPONENT	DESCRIPTION	IMAGE
Controller EDP (LM230WF5-TLF1 1920x1080px 30Pin LCD 23")	HDMI DVI VGA AUDIO LCD Working Board for LVDS LCD Screen Interface. Input power adaptor: 12V DC 2A. It enters standby mode when there is no input signal. Model in stand-by < 1W. Video input: HDMI, DVI, VGA Audio input: HDMI Audio output: jack 3,5mm OSD: contrast brightness, automatic language. Language: english, french, german, spanish, italian. Dimensions: 125 x 58 x 17 mm.	
Kit Raspberry Pi 3 B+	Brand: RASPBERRY PI Series: Raspberry Pi Color: White Weight: 331g Dimensions: 17,6 x 12,2 x 6,4 cm Model: Raspberry Pi 3 B+ OS: Linux CPU: Broadcom BCM2837B0 Speed: 1.4 GHz N. processors: 4 RAM: 1GB HD interface: Ethernet 10/100/1000 GPU: Broadcom Graphics Chipset Type: DDR3 SDRAM Graphic interface: PCI-E Connectivity: Bluetooth, WiFi 802.11ac /b/g/n I/O: 4 USB 2.0	

COMPONENT	DESCRIPTION	IMAGE
Universal Power Ktec - 12V 3A	Brand: POPPSTAR Model: 1004172 Weight: 231 g Dimensions: 12,2 x 7,6 x 5 cm Plug type: EU Power: 24W Input: 100-240V ~ 50/60 hz Output: 12V 3A Cable length: 183 cm Color: black	
HP AIO TOUCH SMART 520 23"	Brand: LG Compatible brands: HP Model: LM230WF5 Dimensions: 23" Technology: LED LCD	

4.5 CONSTRUCTION PHASES

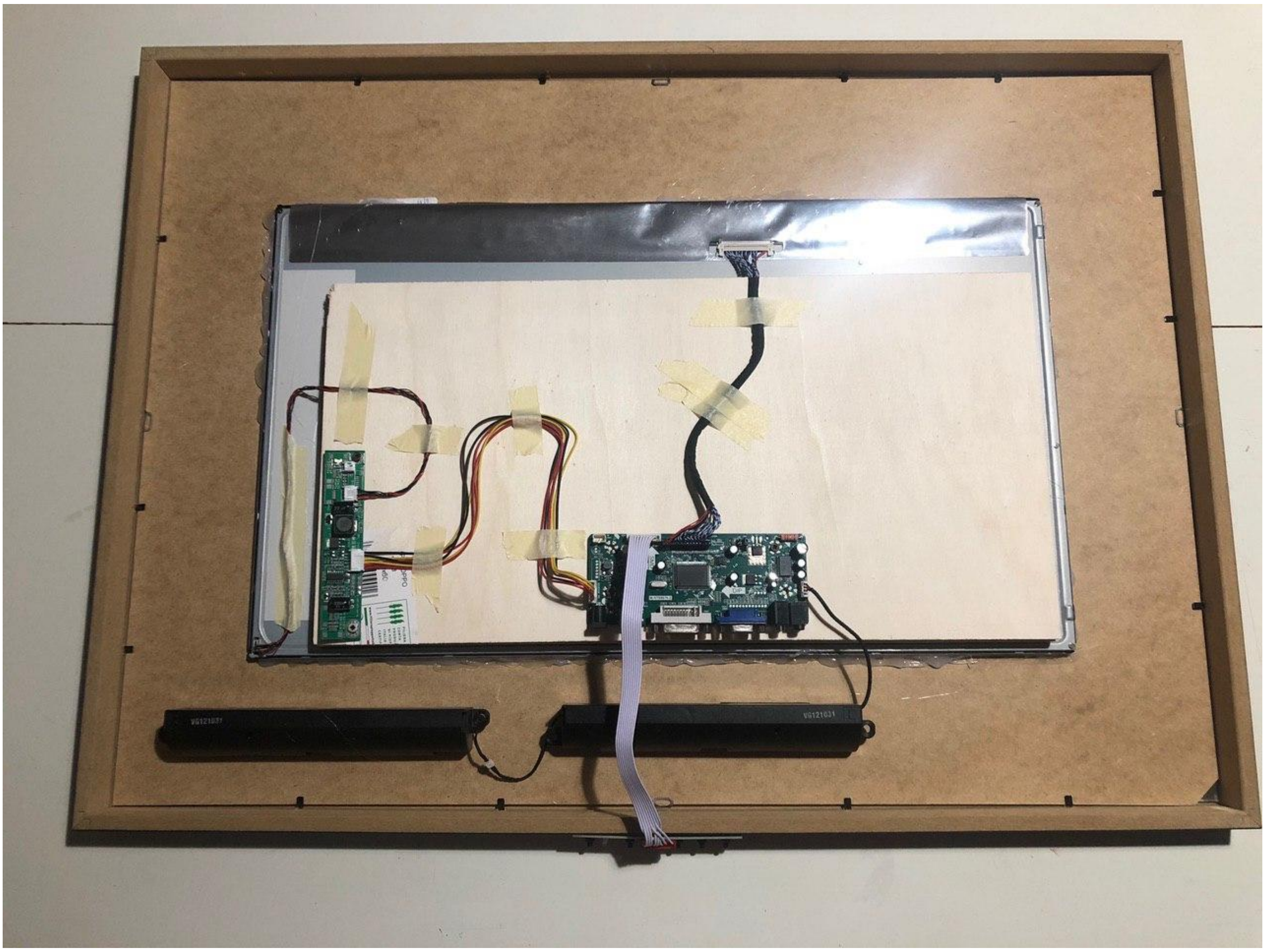
Since the construction of our MinimalMirror involved the use of various different components, to build it correctly and apply the components behind the frame, we followed a precise series of construction phases.

1. Application of the film on the glass
2. Inserting the glass into the frame
3. Cut out the cardboard taking the measurements of the display
4. Insert the card
5. Place the display on top of the cardboard and fix it with hot glue
6. Secure everything with the hooks on the sides of the frame
7. Glue the wooden panel over the display
8. Attach the EDP controller to the panel

4.6 FINAL RESULT

The image on the right shows the back of the mirror once finished assembly.

To secure our EDP controller and avoid interference with the back of our LCD, we screwed the controller itself onto two plywood plates.
LCD, to make it stable, was fixed with hot glue. Finally, we carried out a cable management that ended with the use of paper tape.



5 SOFTWARE

For the software side we relied on MagicMirror². MagicMirror² is an open source modular mirroring platform. With a growing list of installable modules, MagicMirror² allows you to convert a mirror into your personal assistant.
MagicMirror² focuses on a modular plug-in system and uses Electron as an application wrapper. Therefore, no web server or browser installation is required.

MagicMirror² was developed to run on a Raspberry Pi. Electron, the MagicMirror² wrapper application, only supports Raspberry Pi 2, 3 and 4.

The platform was tested on Raspbian. Therefore, once we procured the necessary hardware, we installed the operating system in our Raspberry Pi 3 B+.

5.1 MAGIC MIRROR INSTALLATION AND CONFIGURATION

Once the necessary configurations have been carried out, we proceeded with the manual installation of MagicMirror²:

1. Download and install the latest version of Node.js:

   ```
   curl -sL https://deb.nodesource.com/setup_10.x | sudo -E bash-
   sudo apt install -y nodejs
   ```

2. Cloning of the repository and verification of the master branch:

   ```
   git clone https://github.com/MichMich/MagicMirror
   ```

3. Access to the repository:

   ```
   cd MagicMirror/
   ```

4. Installation of the application:

   ```
   npm install
   ```

5. Copy the config sample file:

   ```
   cp config/config.js.sample config/config.js
   ```

   NOTE: the config.js file allows us to customize our modules by changing the values of the respective parameters.

6. Starting the application:

   ```
   npm run start
   ```

5.2 CHOICE OF MODULES

We then moved on to choosing the modules we wanted to add to our MinimalMirror.

MODULES

- Calendar
- NewsFeed
- OClock
- OpenmapWeather
- NowPlayingOnSpotify
- QRCode
- TelegramBot

5.3 IMPLEMENTATION OF MODULES

Once the application has been configured and tested, to make our *MinimalMirror* more interactive we decided to configure the modules already present and to implement new ones.

CALENDAR

The Calendar module is one of the default MagicMirror modules. This module displays events from an *.ical* calendar.
To use this module, like all the modules described, it must be added to the array in the file `config/config.js`.

```
// DEFAULT CALENDAR WITH OUR ICAL
{
    module: "calendar",
    header: "AGENDA",
    position: "bottom_left",

    config: {
        calendars: [
            {
                symbol: "running",
                url: "https://calendar.google.com/calendar/ical/minimal.mirror.2020%40gmail.com/public/basic.ics" }
        ]
    }
},
```

NEWSFEED
This module displays news headlines based on an RSS feed. The scrolling of news headlines happens based on time (`updateInterval`), but it can also be checked by sending specific notifications in the news feed to the module.

```
//DEFAULT NEWSFEED WITH RSS (Corriere Della Sera)
{
    module: "newsfeed",
    position: "bottom_bar",

    config: {
        feeds: [
            {
                title: "Corriere Della Sera",
                url: "http://xml2.corriereobjects.it/rss/homepage.xml"
            }
        ],
        showSourceTitle: true,
        showPublishDate: true,
        broadcastNewsFeeds: true,
        broadcastNewsUpdates: true
    }
},
```

OCLOCK
This module displays the current year and time. The style is inspired by the Polar Clock.

```
// OCLOCK
{
    module: "MMM-OClock",
    position: "top_left",

    config: {
        locale: "it-IT",
        canvasWidth:300,
        canvasHeight:300,
        centerColor: "#FFFFFF",
        centerR: 50,
        centerTextFormat: "YYYY",
        centerFont: "bold 20px Roboto",
        centerTextColor:"#000000",
        hands: ["second", "minute", "hour"],
        handType: "round",
        handWidth: [25, 25, 25],
        handTextFormat: ["s", "m", "h"],
        handFont: "bold 16px Roboto",
        useNail: true,
        nailSize: 40,
        nailBgColor: "#000000",
        nailTextColor: "#FFFFFF",
```

```
            colorType: "hsv",
            colorTypeStatic: ["red", "orange", "yellow", "green", "blue", "purple"],
            colorTypeRadiation: ["#333333", "red"],
            colorTypeTransform: ["blue", "red"],
            colorTypeHSV: 0.25,
            handConversionMap: {
                "year": "YYYY",
                "month": "M",
                "date": "D",
                "week": "w",
                "day": "e",
                "hour": "h",
                "minute": "m",
                "second": "s"
            },
            secondsUpdateInterval: 1,
            birthYear: false,
            birthMonth: 0,
            lifeExpectancy: 85,
            linearLife: false,
            ageBarColor: [],
            scale: 1,
            canvasStyle: "",
        }
},
```

OPENMAPWEATHER
The OpenmapWeather module is an updated version of the *currentweather* default module. Allows you to view current weather conditions using the Openmap weather API. This module displays the current weather, including wind speed, sunset or sunrise time, temperature and a color icon as an image to view current conditions.

```
// OPENMAPWEATHER
{
    module: "MMM-OpenmapWeather",
    position: "top_right",

    config: {
        location: "Udine",
        locationID: "3165072", //Location ID from http://openweathermap.org/help/city_list.txt
        appid: "52178af66e308000efe51a6d921d6218",  //openweathermap.org API key
        timeFormat: 24,
        onlyTemp: true,
        colorIcon: true
    }
},
```

NOWPLAYINGONSPOTIFY
The module displays the song you are listening to on Spotify.
Shows which device you are playing the song on.
If desired, the user can also view the cover of the current track.

```
// SPOTIFY
{
    module: "MMM-NowPlayingOnSpotify",
    position: "bottom_right",

    config: {
        clientID: "292bd51c55be4068a7ae23ae2d024cb0",
        clientSecret: "52900dc29739435285e9635ce6304f60",
        accessToken: "BQB5c45TlUPPHAoxbRBXH_shId5pI6rYTTKkvK3uk51lXVpYkY9ESEiWa1UtZ8Yw_ah5jeat-FVHB0QK0vh5uF6WOmqHP
        refreshToken: "AQBMHrAaG1p5hdYcq9hTU76XovEH3xrY_xIMi1RuxatlDEDyYT6O32giTBHUyIy7ck0DtGZwIi_UgX5fC3fTBImTfjda
    }
},
```

QRCODE
The following module allows the user to access our GitHub page of the MinimalMirror project via QRCode scan.

```
// QRCODE
{
    module: 'MMM-QRCode',
    position: "bottom_right",

    config: {}
},
```

TELEGRAMBOT
Thanks to this module, the user has the ability to remotely control the MinimalMirror via Telegram thanks to the commands made available by our bot (MinimalMirror_bot).

```
// TELEGRAM BOT
{
    module: 'MMM-TelegramBot',
    position: "top_left",

    config: {
        telegramAPIKey : '1303565388:AAF36v-OgAs8tfud9pIUUYV-p-j6PbYTJds',
        allowedUser : ['kouatemuhamed'],
        adminChatId: 622250826,
        favourites:["/commands", "/modules", "/hideall", "/showall"],
        telecast: true,
        telecastLife: 1000 * 60, //1 minute = 1000ms x 60
        telecastLimit: 10,
    }
},
```

6 CONCLUSIONS

The design of the mirror and the implementation of the modules allowed us to expand our professional knowledge, putting ourselves to the test, with all the difficulties of the case. The team had to face many difficulties in finding the necessary hardware, due to the unfortunate COVID-19 virus that has affected the whole world.
However, this impediment has taught us to work despite external adversities.

Everyone had an active and important role in the creation of the project,
comparing and working in a group.

The product we made has exceeded initial expectations. The commitment and dedication dedicated to this project has allowed us to achieve a truly satisfactory result.

For more information about the project, you can visit the repository on Github:

https://github.com/KouateMuhamed/MinimalMirror